

TEMPLATE PATTERN

Behavioural Design Pattern

Problem Statement

Imagine you're building a system that **reads data** from different file types like **CSV**, **XML**, and **JSON**. Every file reader needs to follow the same basic process: **open** the file, **read** the content, and **close** the file.

If you don't use the Template Method Pattern, you'll end up writing the same opening and closing logic over and over again in each file reader class.

Problems in our code

Repeated Code: The `openFile()` and `closeFile()` methods are written again in every single file reader class, creating unnecessary duplication.

Hard to Maintain: If you need to modify the file opening or closing logic, you have to update it in all reader classes separately. This breaks the **DRY (Don't Repeat Yourself)** principle.

UML Diagram

